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Education

- 2021 – Now 📖 **Ph.D. University of Delaware**; Computer & Information Sciences.
Advisor: *Christopher Rasmussen*.
- 2018 – 2020 📖 **M.S. Boston University**; Computer Science.
Advisor: *Margrit Betke*.
- 2014 – 2018 📖 **B.Eng. Wuhan University of Technology**; Software Engineering.
GPA: 91.04/100; Rank: 2/228.

Research Publications

Journal & Conference & Workshop

- 1 **Q. Wang**, H. Dai, H. Zhang, C. Rasmussen, and B. Wang.
“When One Modality Rules Them All: Backdoor Modality Collapse in Multimodal Diffusion Models.”
Workshops of the International Conference on Learning Representations (ICLR-W), 2026.
▷ The first two authors contributed equally to this work.
- 2 **Q. Wang**, F. Du, P. Maneriker, J. Jin, and C. Rasmussen.
“Toward Low-Latency Vision-Language Models with Doubly-Correct Predictions in Egocentric Visual Understanding.”
International Conference on Intelligent Robots and Systems (IROS), 2026.
▷ Acceptance rate 36%; Top conference in Robotics.
▷ Work done during an internship at Dolby Laboratories Inc.
- 3 **Q. Wang**, Y. Liang, M. Li, T. Zhou, and C. Rasmussen.
“History-Conditioned Spatio-Temporal Visual Token Pruning for Efficient Vision-Language Navigation.”
International Conference on Intelligent Robots and Systems (IROS), 2026.
▷ Acceptance rate 36%; Top conference in Robotics.
- 4 **Q. Wang**, E. Chinkaka, R. Richaud, M. Haghdadi, C. Wolk, K. V. Oromeng, K. F. Davis, F. Bianco, X. Peng, and J. M. Klinger.
“MO-SAM: Testing the reliability and limits of mine feature delineation using Segment Anything Model to democratize mine observation and research.”
PLOS Sustainability and Transformation, 2025.
- 5 **Q. Wang**, T. Li, K. X. Nguyen, and X. Peng
“Beyond Accuracy: On the Effects of Fine-tuning Towards Vision-Language Model’s Prediction Rationality.”
Association for the Advancement of Artificial Intelligence (AAAI), 2025.
▷ Acceptance rate 23.4%; Top conference in Artificial Intelligence.
- 6 **Q. Wang**, L. Zhao, L. Yuan, T. Liu, and X. Peng
“Learning from Semantic Alignment between Unpaired Multiviews for Egocentric Video Recognition.”
International Conference on Computer Vision (ICCV), 2023.
▷ Acceptance rate 26.1%; Top conference in Computer Vision & Pattern Recognition.
- 7 **Q. Wang**, B. Fu, M. Li, J. He, X. Peng, and Y. Qiao
“Region-aware Arbitrary-shaped Text Detection with Progressive Fusion.”
IEEE Transactions on Multimedia (TMM), 2022.
▷ Impact factor 7.3; Top journal in Multimedia.

▷ The first two authors contributed equally to this work.

- 8 Y. Zou, J. Choi, **Q. Wang**, and J.-B. Huang
“Learning representational invariances for data-efficient action recognition.”
Computer Vision and Image Understanding (CVIU), 2022.
▷ Impact factor 4.5; Prestigious journal in Computer Vision & Pattern Recognition.
- 9 Y. Zheng, **Q. Wang**, and M. Betke
“Semantic-Based Sentence Recognition in Images Using Bimodal Deep Learning.”
IEEE International Conference on Image Processing (ICIP), 2021.
▷ Prestigious conference in Computer Vision & Pattern Recognition.
- 10 **Q. Wang**, Y. Zheng, and M. Betke
“A Method for Detecting Text of Arbitrary Shapes in Natural Scenes That Improves Text Spotting.”
Workshops of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR-W), 2020.

Internship & Collaboration

May 2026 – Aug 2026	📌 PhD Research Intern , Dolby Laboratories Inc.
Sep 2021 – Now	📌 Graduate Research Assistant , University of Delaware.
Jun 2025 – Sep 2025	📌 PhD Research Intern , Dolby Laboratories Inc.
Sep 2021 – Nov 2022	📌 Research Collaboration , Google Research.
Jun 2021 – Aug 2021	📌 Applied Scientist Intern , Amazon Web Services (AWS).
May 2020 – Aug 2020	📌 Visiting Student , Shenzhen Institute of Advanced Technology.

Selected Projects

Oct. 2025 - Mar. 2026	📌 Dynamic Vision Lab, University of Delaware ; Introduced a vision token pruning strategy to improve efficiency in Vision-Language Navigation for Vision-Language-Action models; <u>work accepted by IROS-2026</u> .
Oct. 2025 - Jan. 2026	📌 Dynamic Vision Lab, University of Delaware ; Investigated modality dominance and imbalance issues in diffusion models for visual data generation; <u>work accepted by ICLR-W-2026</u> .
Jun. 2025 - Sep. 2025	📌 Dolby Laboratories Inc. ; Introduced a new approach that improves both the reliability and efficiency of Vision-Language Models (VLMs) for video learning and understanding; <u>work accepted by IROS-2026</u> .
May. 2024 - May. 2025	📌 Deep-REAL Lab, University of Delaware ; Introduced a new dataset for video understanding and benchmarked various models, including Multimodal Large Language Models such as Video-LLaVA and VideoLLaMA2.
Sep. 2023 - Aug. 2024	📌 Deep-REAL Lab, University of Delaware ; Designed experiments to investigate the impact of fine-tuning on the prediction rationality of Vision-Language Models (e.g., CLIP); <u>work accepted by AAAI-2025</u> .
Mar. 2023 - Apr. 2024	📌 Dept. of Geog., University of Delaware ; Proposed a method utilizing the “Segment Anything Model (SAM)” to detect mine features in satellite imagery; <u>work accepted by PLOS Sustainability and Transformation, 2025</u> .
Sep. 2021 - Nov. 2022	📌 Google Research (Collaboration) ; Developed a method to align unpaired multiview videos with varying cross-view semantic information, utilizing the capabilities of Large Language Models; <u>work accepted by ICCV-2023</u> .
May. 2021 - Aug. 2021	📌 Amazon Web Services ; Developed a method for image-text retrieval.
May. 2020 - Nov. 2020	📌 Shenzhen Inst. of Adv. Tech., Chinese Academy of Sciences ; Developed a new method for scene text detection; <u>work accepted by TMM-2022</u> .

Awards

- 2024  **Outstanding Conference Travel Award**, CIS department of University of Delaware.
  **CIS Distinguished Graduate Student Award**, University of Delaware.
- 2023  **Outstanding Conference Travel Award**, CIS department of University of Delaware.

Services

- Conference Reviewer  NeurIPS 2026, ICML 2026, CVPR 2025-2026, ICCV 2025, ECCV 2024-2026, ACMMM 2025, ACMMM 2025 Datasets, BMVC 2024-2026.
- Journal Reviewer  IEEE TPAMI, IEEE TIP, IEEE TMM, Artificial Intelligence Review, ACM Computing Surveys, IEEE Access, Journal of Supercomputing, PLOS ONE.
- Program Committee  AAAI 2026-2027, ACMMM 2026.
- Vol. Conf. Reviewer  CVPR 2023, NeurIPS 2023, AAAI 2024, ICLR 2025.
- Vol. Jour. Reviewer  IEEE TAI, IEEE TPAMI, ACM TIST.

Invited Talk

- Sep. 2023  **Extraction, Effluent, and Enumeration in Extraglobal Geopolitics**; Off-Earth Geopolitics Workshop. University of Oxford (virtually).

Skills

- Coding  Python, PyTorch, $\text{\LaTeX}$, ...
- Large Models  CLIP, SAM, BERT, VLA, Diffusion Model, Video-MLLM.....